

Entangled State

QC@IITI

Introduction

In modern literature, the term *Entanglement* is used quite often. Here, we shall try to understand what it actually means in the context of Quantum Computing, in a relatively simple way. We will also try to understand where we use it, why we use it, and also how we produce it.

Understanding Entanglement

Defining Entanglement

Formally, Entanglement in the context of Quantum Computing is defined as follows: *Entanglement is a fundamental concept of quantum mechanics that describes a non-classical correlation, or shared quantum state, between two or more quantum systems (or quantum particles) even if they are separated by a large distance.*

Now that we know the formal definition, let's start to unpack the definition. The first thing that we must know is that Entanglement is a *Quantum Mechanical* term, meaning that it is an effect completely disconnected from classical terms, and we must think differently to understand this.

Secondly, Entanglement is an effect that occurs between multiple bodies/particles, that is to say, it is a relationship, where information about one body gives us a different packet of information about another body in entanglement with the first body.

Lastly, Entanglement is not restricted by distance. You may have the two bodies be millions if not billions of miles apart, but they may still be in an Entangled state.

Effects of Entanglement

Here, we will switch to basic Dirac notation to first understand this mathematically. Let's say, that we have a Unitary Operator U_1 , which takes in the two long bit-string $|00\rangle$, and outputs the state $|\psi\rangle$. It is not important what U_1 is, it's just important to know that such a quantum gate exists, and it produces the following:

$$U_1 |00\rangle = |\psi\rangle = \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |10\rangle$$

Now, we say that the two qubits representing the state are in Entanglement. What does that mean? Well according to the definition, by knowing the information about one particle, we should be able to obtain information about the other particle. Let's take the following example. Say, we separated these two qubits to two different rooms. A person, say Alice, reads the qubit in room 1, and finds that the reading is 0. Since we know that the state is either a $|01\rangle$ or a $|10\rangle$, then by knowing that we have found a 0 bit, we know that upon measuring that other qubit in the other room, we will certainly obtain a 1. In this way, by measuring a single particle of an entangled pair, we were able to collapse the state of the other qubit onto a state without even reading it.

Here, it is important to keep in mind the distinction between quantum mechanical entanglement and classical causality. The example that was given is different in nature from causality. Let's understand it with another example. If Child A was given two balls, a red one and a green one, was told to close his eyes, and then the two balls were hidden in two different boxes, then when the Child opens a box, and finds that it was a green ball, he will definitely know that the other box contains a red ball, yet this is not entanglement.

The Difference here lies in the inherent uncertainty. In the first example, if Alice was to read her qubit, the "ball inside the box" so to say in this case, does not have a value defined with certainty, as compared to the other example. If theoretically, we were able to make 100 states just exactly like this, and separated the qubits in the exact same way, and had Alice open the exactly same box 100 times, still we would have her read roughly half of them as 1 and half of them as 0. This is because the value of the qubit isn't intrinsically defined until after it is actually observed. In all these cases still, we have determined the state of the other qubit without opening it, thus showing that the two qubits were entangled.

How do we Entangle stuff

Now that we know what Entanglement does to a state, but now we will find how we can put multiple states into entanglement in the first state, at least, thru our imaginary operator U_1 . Logically speaking, let's presume that we have access to two qubits, let's name them Q_1 and Q_2 , both initialised to the $|0\rangle$ state. If we want to make the operator U_1 , we will follow the following process:

First, we apply a basic X gate (Functions like a NOT gate in classical computing), on Q_2 . This will result in us obtaining the $|01\rangle$ state for the system Q_1Q_2 .

Next, we apply a Hadamard gate to the qubit Q_1 , making the net system $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$.

Lastly, we apply a CNOT gate (Controlled Not), with Q_1 as the control and Q_2 as the target. This will turn the system into $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. This will produce the state that we desired.

So, we can say that U_1 can be produced by the following way:

$$U_1|00\rangle = CNOT_{12}(H \otimes I)(I \otimes X)|00\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$$

Uses of Entanglement

Entanglement is used in every single way you can imagine within Quantum Computing. One specific beginner level use case is in the quantum teleportation protocol, where a sender (Alice) transmits a qubit to a receiver (Bob) by making use of a shared entangled quantum state (one e-bit, to be specific) along with two bits of classical communication. You can read more about it if you are interested [here](#). Some other places where we can see Entanglement in action is [Superdense Coding](#) and the [CHSH Game](#). You can study about these topics if they interest you.

Lastly, you can learn more about entanglement [here](#)